

# PRERAK NAIN

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AI/ML Engineer • LLM Agents & RAG • Full-Stack • Production Systems, Shipped Solo

B.Tech Computer Science student who builds and ships production AI systems solo, end to end — ML models, LLM/agent pipelines, backend APIs, databases, and deployed frontends. Comfortable across the full stack and the full AI lifecycle: research, model training, calibration, production debugging, and turning technical work into something real users and clients actually use.

## EDUCATION

**B.Tech, Computer Science and Engineering — Bennett University, Greater Noida**

Aug 2023 — May 2027

CGPA: 7.4/10

## EXPERIENCE

**DevRel & Forward Deployed Engineer Intern — Metabees (Keirolabs) — Remote**

Jul 2026 — Present

- Serve as the technical bridge between engineering and go-to-market for Keirolabs' production search API — turning core technical capability into developer-facing content, positioning, and integration support.
- Act as forward-deployed engineer on live customer integrations, closing the loop between real-world field feedback and product decisions.
- Ran benchmark testing on the search API and built a status page for monitoring live system uptime and health.

**Independent AI Automation Consultant — Freelance, Remote**

2025 — Present

- Design and deploy applied AI/automation systems for remote SMB clients end-to-end: data pipeline, model/workflow build, containerized deployment, and monitoring.

## RELEVANT PROJECTS

**Aegis AI — Production AI Security Platform, Built & Shipped Solo**

Python, PyTorch, LangGraph, FastAPI, PostgreSQL (Neon), Next.js, React, TypeScript, Clerk, Docker — live at [aegis-ai-frontend-tau.vercel.app](https://aegis-ai-frontend-tau.vercel.app)

- Built and deployed a 4-model ML ensemble (99.88% accuracy, 13 attack classes, 2.8M+ real network flows) feeding a LangGraph agent that grounds severity assessment and recommendations in a RAG knowledge base, turning raw detections into audited, human-readable threat reports.
- Found and fixed three separate model failures the headline accuracy number was hiding: a misconfigured LightGBM training at 52% (30-trial Optuna tuning → 99.96%), 0% recall on a rare imbalanced attack class (SMOTE + WGAN-GP synthetic augmentation → 100% recall, raising macro F1 90.69% → 98.43%), and a 32.7-point gap between stated confidence and actual correctness (Platt Scaling calibration → 0.6pp).
- Diagnosed and fixed a hard 512MB memory ceiling crashing the service in production on Render — redesigned RAG retrieval from a Chroma + neural-embeddings store down to TF-IDF, cutting memory footprint to fit the constraint with no loss in retrieval quality on the target knowledge base.
- Owned the full stack solo: a multi-tenant FastAPI backend (per-customer API-key auth, rate limiting, locked-down CORS) on Neon Postgres, a Next.js/React/TypeScript frontend on Vercel with Clerk authentication, and a signature-verified (Svix) webhook that auto-provisions an isolated customer account on signup.

**SCAM Framework — Explainable AI Research**

IEEE-format, co-authored

- Co-authored a 6-page paper on a comparative model-attribution framework, contributing experimental design and evaluation methodology for explainable AI outputs.

## TECHNICAL SKILLS

**AI / ML / LLM:** Python, PyTorch, scikit-learn, XGBoost, LightGBM, LangGraph, RAG (TF-IDF & embedding-based retrieval), prompt engineering, HuggingFace transformers, Optuna, model calibration (Platt Scaling)

**Engineering:** FastAPI, PostgreSQL (Neon), REST API design, multi-tenant auth, rate limiting, Docker, Git/GitHub, production deployment & debugging (Render, Vercel)

**Frontend:** Next.js, React, JavaScript, TypeScript, Clerk auth

## ACHIEVEMENTS & CERTIFICATIONS

- 3rd Place, National-level Quiz organised by ISRO.
- Machine Learning coursework certification, University of Illinois Urbana-Champaign (Coursera).